

ORIGINAL ARTICLE

Correlation Analysis of the Optics of Progressive Addition Lenses

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ABSTRACT: *Purpose.* To investigate the relations between selected key optical parameters and the sizes of the clear viewing areas of progressive addition lenses (PALs). *Methods.* The optics of 28 PALs (plano with +2.00 D add) currently on the market were measured with a Rotlex Class Plus lens analyzer. Horizontal cross sections were analyzed in 1 mm vertical steps with respect to the fitting cross. Distance, intermediate, and near viewing zone widths and areas were calculated from the measurements. The maximum amount of unwanted astigmatism, minimum zone width (0.50 DC limit), and maximum power rate in the corridor were also recorded for each lens. Correlation coefficients were determined for all relations. *Results.* Each of the three viewing zone areas had a significant negative relation with the other (r of -0.4 to -0.8), indicating design tradeoff. Maximum power rate was significantly related to minimum zone width ($r = -0.695$), which was significantly related to maximum astigmatism ($r = -0.616$), but there was not a significant relation between maximum power rate and maximum astigmatism. Higher power rates and narrower minimum zones were significantly related to smaller intermediate and larger near zones ($r = 0.4$ to 0.9). Maximum astigmatism was related to distance zone width ($r = 0.42$) and to intermediate zone size ($r = -0.4$ to -0.56), but not significantly related to near viewing zone. Power rate and astigmatism each vary relatively uniformly across each lens. *Conclusions.* The fundamental relation appears to be between power rate and zone width, each of which is highly related to sizes of the intermediate and near viewing zones. The maximum amount of astigmatism is related to zone width, but not to maximum power rate. The amount of astigmatism is unrelated to the size of the near zone. The pattern of correlations between the optical and viewing zone parameters help identify the underlying optical relations of PALs. (*Optom Vis Sci* 2004;81:350-361)

Key Words: multifocals, ophthalmic lenses, ophthalmic optics, optics, presbyopia, progressive addition lenses, vision

Progressive addition lenses (PALs) are seamless multifocal lenses characterized by having a location in the superior portion of the lens that contains the distance power, a location in the lower portion of the lens that contains the near power, and a continual smooth change of power between the two. PALs have proven to be a very successful treatment for presbyopia.¹⁻³

For most PALs, the progressive optics are located on the front surface with a spherical or toric back surface to provide the appropriate prescription. The discussions in this article assume this arrangement. Although some designs use both surfaces to create the progressive characteristics, the principals and results in this study apply regardless.

As result of the progressive design, PALs also have unwanted astigmatism peripheral to the power change corridor. Such astigmatism is usually located in the lower quadrants of the lens. To accomplish the power progression, the radius of curvature must gradually decrease proceeding from the distance center to the near center. The result is a considerably steeper curvature in the bottom

of the lens compared with the top. For the power to be spherical, the radius of curvature must change in the vertical and horizontal meridians. With these constraints, it becomes necessary to deviate from spherical surfaces in the lens periphery to make a seamless lens. The deviations result in unwanted astigmatism. The amount of astigmatism is inevitably related to rate of change of power. Von Minkwicz,⁴ in a theoretical analysis of a progressive surface, suggested that the unwanted astigmatism at points outside the corridor will grow at twice the rate of increase of power along the corridor.

The lens designer has numerous options in the design of a particular PAL, such as rate of power progression, distance between the distance and near centers, the width of the error-free viewing zones, and magnitude of unwanted astigmatism. As result, there is almost no limit to the number of possible PAL designs; hence, numerous different designs exist in the marketplace. As would be expected, there are tradeoffs in the designs. It is not possible to optimize the sizes of the distance, intermediate and near zones,

have a short corridor from distance to near, and also have a lens with minimal unwanted astigmatism. The interdependencies of the design variables and the viewing zones are the subject of the current study.

Charman⁵ determined that the amount of unwanted astigmatism is related to the power of the add, a reasonable relation based on the fact that a higher add requires greater change in curvature, which results in greater magnitude of unwanted astigmatism. Simonet et al.⁶ studied two different PAL designs and determined that the ratio of astigmatism to the add was fairly constant within the design but that the ratio was very different between designs (i.e., different designs had a different ratio). They also noted that astigmatism seemed proportional to the width of the near zone and inversely proportional to corridor length. These findings support the common categories of “hard” and “soft” designs. Hard designs characteristically have wider viewing zones, a shorter distance between the distance and near poles, and higher magnitudes of unwanted astigmatism, whereas soft designs are the opposite. However, PAL design is considerably more complex than this simple categorization. Other previous studies have likewise shown large differences across different lens designs in terms of viewing zone sizes and unwanted astigmatism.^{7, 8}

In a recent study, Sheedy⁹ discussed the systematic optical measurements of 28 different PAL designs. Several features of the lenses, including sizes of the distance, intermediate, and near zones, magnitude of unwanted cylinder, and height of the add, were measured and reported. The optical features of the lenses were presented and analyzed as they likely apply to individual patient needs. The objective of this current study was to further analyze the optical data on the 28 lens designs for the purposes of identifying the tradeoffs and relations between the optical features of PALs.

METHODS

The measurement methods have been reported elsewhere⁹ and are summarized herein. Twenty-eight PALs of different designs were selected for measurement. All PAL designs were intended for general use and included some designed as short corridor lenses for small frames. All were right lenses and ordered to have plano distance power with +2.00 D add.

Lens contour plots were measured with a Rotlex Class Plus lens analyzer. This instrument operates as a moiré deflectometer using a point source rather than a collimated beam. Light diverging from a laser point source is incident on the back surface of the tested lens. The location of the laser point source is variable and depends on the power of the lens. For lenses of plano power (in this study power ranged from plano to +2.00 D) it is 92 mm; for +10 D, it is 50 mm. Ideally the laser source would be located to simulate the center of ocular rotation (about 27 mm). However, previous research has shown that differences in this measurement parameter have no significant effect on measurements for lenses of low power.^{10, 11} Therefore, this factor is considered nonsignificant in this study. The rays are refracted by the lens and pass through two gratings to form a moiré pattern on a diffusive screen. Proprietary algorithms process the fringe data in the image to create arrays of local wavefront properties, in particular, the two principal curvatures and axis directions. The arrays are used to calculate two-dimensional maps of local power, cylinder, and axis.

All PALs were measured with the prism reference line markings appropriately aligned in the instrument and a data file saved. The location of the fitting cross as specified by the manufacturer with respect to the 0 to 180 line was identified and used as the center of the *xy* coordinate system for all lenses. Hence, all lens measurements were specified with respect to the location intended to be fitted before the pupil. All measurements were determined with the “DST” mode of the instrument; that is, all the measures on each lens were normalized to an assigned power of plano at the manufacturer-specified distance location, thereby mitigating the effects of laboratory surfacing variances.

Data were acquired from each file in a stepwise manner by examining the data files in 1 mm vertical increments beginning at 10 mm above the fitting cross ($y = -10$) and extending to 25 mm below the fitting cross ($y = +25$). The following data points were recorded at each *y* value: the left and right *x* coordinates of the first 0.50 D cylinder (DC), first +0.25 D sphere (DS) (distance area only), first +0.50 D sphere (distance area only), first +1.75 D (near area only), and first +2.00 D (near area only), value of the greatest amount of unwanted cylinder at that level, and spherical power (maximum plus power) in the center of the corridor. The data files also were analyzed to provide the following data for each 0.25 D increment of power along the center of the corridor: *y* location, left and right *x* values of 0.50 D cylinder limits, and maximum unwanted cylinder at that level. In data analysis, unwanted cylinder of 0.50 D was used as a limit of zone width.

For analysis, parameters relating to the distance, intermediate, and near viewing zones were derived from the data files for each lens. A schematic illustrating several of the parameters is shown in Fig. 1. For the distance viewing zone, unwanted refractive error of 0.25 DS or 0.50 DC, whichever was more limiting, constituted the lateral limits of the usable distance viewing zone. The zone widths at -1 , 0 , and 1 were used as distance zone parameters. In addition, the area of the distance zone was calculated by summing the zone widths from 1 mm above the fitting cross down to the lowest level of the distance zone (<0.25 D add) for each lens. This effectively integrated the area in steps of 1 mm. The width at each 1 mm step therefore represents the area extending 0.5 mm above and below it. Hence, the calculated distance zone represented the area from 1.5 mm above the fitting cross to the lowest extent of the distance zone. The widths of the intermediate zone (0.50 DC limits for add powers of +0.75, +1.00, and +1.25 D) were used as intermediate zone parameters. The area of the intermediate viewing zone was calculated by integrating the area from +0.75 to +1.50 D add in 0.25 D increments. Although all lenses had a nominal near add power of +2.00 D, 12 of the 28 designs did not progress entirely to +2.00 D; hence, +1.75 D or greater and less than 0.50 DC, whichever was more limiting, was used to limit the extent of the near viewing zone. The level of first appearance (descending from the fitting cross) of +1.75 D and the near zone widths at 16 and 20 mm were used as near zone parameters. The integrated near zone down to 16.5 and 20.5 mm were also used as near viewing zone metrics.

Three global design parameters were also identified for each lens. These optical limit values were maximum power rate, minimum zone width, and maximum astigmatism. The rate of power change (D/mm) at each 1 mm level in the center of the corridor was calculated by determining the difference of power between the

levels that were 1 mm above and below that level and then dividing by 2. The maximum power rate on each lens was identified along with the following associated variables: the y level at which it occurred, the maximum astigmatism at that level, the zone width at that level, and the add power at that level. Similarly, the minimum zone width (0.50 DC limits) on the lens was determined along with its y level, and the maximum astigmatism, power rate, and add power at that level. Also, the maximum unwanted astigmatism was identified along with its y location, the power rate, zone width, and add power at that level.

Data were analyzed by calculating correlation coefficients¹² for relations of the viewing zone and optical limit value parameters across the 28 lenses.

RESULTS

The mean values for the viewing zone parameters are illustrated in Fig. 1.

Relations of Vision Zone Measurements to One Another

The relations of each vision zone measurements to each other vision zone measurement are shown in Table 1.

The interzone measurements are all significantly ($p < 0.001$) related to one another; all the distance measures are highly significantly related to all other distance measures, intermediate to intermediate, and near to near. The Pearson r values are all quite

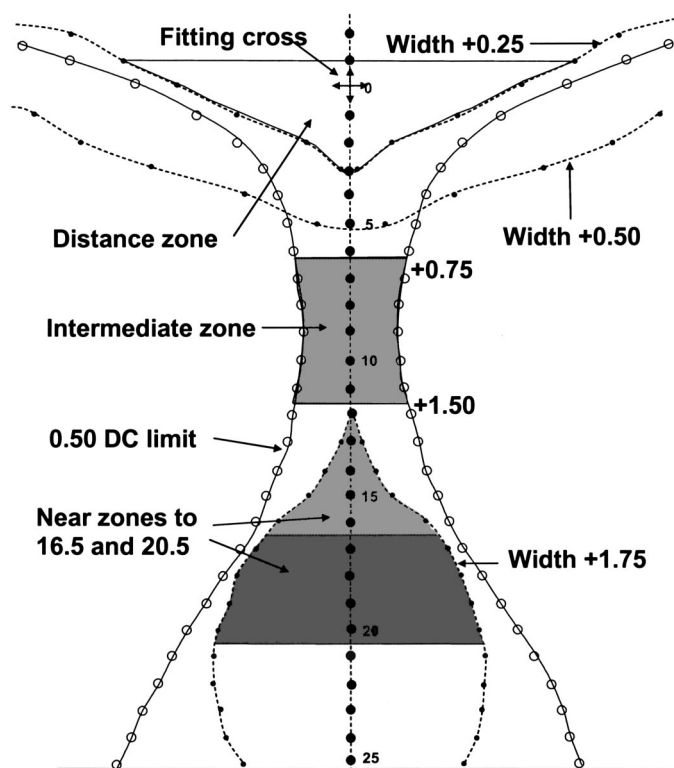


FIGURE 1.

An illustration of some of the viewing zone parameters and zone limits used in this study. The drawing is scaled to the mean values obtained for the 28 lenses.

large, mostly 0.8 and higher. These findings show considerable co-variation of the measures within a zone for each lens. This result is not surprising; it is to be expected that the various widths and area measures within the same viewing zone for a given lens would be highly related to one another.

Distance Zone and Intermediate Zone Relations

Each distance width and area measurement has a negative correlation with each intermediate width and area measurement (Table 1). This is a clear indication of design tradeoff; that is, larger distance zone widths and areas generally are associated with narrower intermediate widths and areas and vice versa. However, only a few of the individual relations approach statistical significance. The greatest negative correlation (-0.409) and most statistically significant comparison between measurements in these two categories is between the distance width at 1 mm and the intermediate width at $+0.75$ D; these represent the lowest width measure in the distance zone and the highest width measure in the intermediate zone, respectively. However, considerable variation from an inverse relation exists even between these two measures ($R^2 = 0.167$, meaning that the variance in one explains only 16.7% of the variance in the other). Although there is tradeoff (inverse relation) between distance and intermediate measures, the considerable individual variation indicates that individual lens design can mitigate the tradeoff. The weakest negative correlation between the distance and intermediate measures was between the distance and intermediate areas ($r = -0.096$, $p = 0.627$). Although there is a general negative correlation between distance and intermediate width and area measurements, it is not rigid, and considerable variation exists across lenses.

Distance Zone and Near Zone Relations

Each of the individual distance width and area measures has a negative correlation with each of the near width and area measurements, similar to the relation between distance and intermediate measures. This indicates a general design tradeoff between distance zone widths and near zone widths. The greatest and most significant relation ($r = -0.431$, $p = 0.022$) is for the distance area (-1 to $+3$) and the near zone width at 20 mm. There is still a great deal of variance between the two; r^2 is only 0.186.

A positive relation between each of the distance measures and the first (highest level) occurrence of $+1.75$ D add can be seen in Table 1. This means that greater distance width and area is associated with a lower occurrence (higher y value) of the first appearance of the full add (taken to be $+1.75$ D). In practical terms, this represents a design tradeoff between distance width and area and the height at which the full add is attained. A design with greater distance width and area will generally require the wearer to depress their eyes more to obtain the near add. A lower add would also necessitate a greater fitting height in the frame.

It is noteworthy that each distance width and area measure bears a stronger relation to the near width and area at 20 and 20.5 than to the near width and area at 16 and 16.5. Similarly, the relation to the first occurrence of the $+1.75$ D add is strongest for the distance

TABLE 1.
Correlation coefficients of viewing zones compared with viewing zones

	Distance				Intermediate				Near				
	Width −1	Width 0	Width 1	Area −1 to +3	Width 0.75	Width 1.00	Width 1.25	Area 0.75 to 1.50	First 1.75	Width 16	Area 16.5	Width 20	Area 20.5
Distance													
Width −1													
Pearson r	1	0.946	0.794	0.892	−0.361	−0.399	−0.207	−0.109	0.393	−0.253	−0.362	−0.323	−0.349
Significance		<0.001	<0.001	<0.001	0.059	0.035	0.29	0.58	0.038	0.194	0.058	0.094	0.069
Width 0													
Pearson r		1	0.883	0.956	−0.357	−0.373	−0.17	−0.153	0.368	−0.266	−0.316	−0.369	−0.342
Significance			<0.001	<0.001	0.062	0.051	0.387	0.436	0.054	0.172	0.101	0.053	0.075
Width 1													
Pearson r			1	0.915	−0.409	−0.334	−0.198	−0.196	0.282	−0.26	−0.226	−0.287	−0.236
Significance				<0.001	0.031	0.082	0.313	0.318	0.147	0.181	0.248	0.139	0.226
Area −1 to +3													
Pearson r				1	−0.316	−0.266	−0.096	−0.092	0.42	−0.28	−0.311	−0.431	−0.363
Significance					0.102	0.171	0.627	0.641	0.026	0.149	0.108	0.022	0.057
Intermediate													
Width 0.75													
Pearson r					1	0.897	0.822	0.827	0.479	−0.536	−0.55	−0.419	−0.569
Significance						<0.001	<0.001	<0.001	0.01	0.003	0.002	0.027	0.002
Width 1.00													
Pearson r						1	0.891	0.851	0.493	−0.481	−0.484	−0.416	−0.516
Significance							<0.001	<0.001	0.008	0.01	0.009	0.028	0.005
Width 1.25													
Pearson r							1	0.9	0.619	−0.58	−0.547	−0.541	−0.613
Significance								<0.001	<0.001	0.001	0.003	0.003	0.001
Area 0.75 to 1.50													
Pearson r								1	0.796	−0.688	−0.768	−0.607	−0.78
Significance									<0.001	<0.001	<0.001	0.001	<0.001
Near													
First 1.75													
Pearson r									1	−0.727	−0.91	−0.763	−0.922
Significance										<0.001	<0.001	<0.001	<0.001
Width 16													
Pearson r										1	0.727	0.702	0.83
Significance											<0.001	<0.001	<0.001
Area 16.5													
Pearson r											1	0.685	0.931
Significance												<0.001	<0.001
Width 20													
Pearson r												1	0.885
Significance													<0.001
Area 20.5													
Pearson r													1
Significance													

width at -1 and least for $+1$. These relations show a tendency for the upper portions of the distance area to be most related to the lower portions of the near zone.

Intermediate Zone and Near Zone Relations

Of the three pair-wise comparisons of the three zones, this comparison of zone measurements clearly produces the strongest and most significant relations. Each intermediate zone width and area is negatively and significantly related to each near zone width and area; that is, wider and larger zones in one are associated with narrower and smaller zones in the other. The greatest negative correlations are between the intermediate calculated area and the two near calculated areas to 16.5 and 20.5 mm.

There is also a strong positive and statistically significant relation between the intermediate area and first occurrence of the $+1.75$ D add. A greater intermediate area is strongly associated with a lower (higher y) occurrence of the add.

Relations Between the Viewing Zone Measurements and the Optical Limit Measures

The correlation coefficients and significance values for the comparisons of the viewing zone measures with the optical limit values (maximum astigmatism, maximum power rate, and minimum zone width) are shown in Table 2. The mean values and locations of the optical limit values for the 28 lenses are shown in Table 3.

Distance Zone Relations with Optical Limit Values

The most significant relation of distance measures to the three optical limits values was a positive relation between the distance zone width at -1 (the highest level) with the amount of astigmatism at the level of greatest power rate change ($r = 0.502$, $p = 0.006$). However, there are also similar significant relations between the distance width at -1 with the other two measures of astigmatism: the maximum astigmatism on the lens ($r = 0.483$, $p = 0.013$) and the amount of astigmatism at the level of the minimum zone width ($r = 0.435$, $p = 0.021$). It is noteworthy that the relations to maximum astigmatism are not nearly as strong for the distance widths at 0 and $+1$, indicating that the effects of maximum astigmatism on distance zone are stronger near the upper part of the distance zone instead of the lower part. Clearly, higher magnitudes of unwanted astigmatism are associated with wider distance zones, especially near the upper portion (-1) of the distance zone as defined in this study.

The other interesting and significant relations between distance zone measurements and optical limit values is the negative relation between the add power at the location of the minimum zone width and the distance width and area. In other words, greater distance zone width and area are associated with lenses in which the add power at the minimum zone width is less. Another way to state this is that lenses tend to have wider distance zones if the add progression occurs lower relative to the minimum zone width. Again, the greatest correlation is for the highest distance zone width at -1 .

Positive relations exist between the distance width and area and the y value at which the maximum power rate occurs. This makes

sense because the lower the level of maximum rate change is, the wider the distance zone at the fitting cross.

Intermediate Zone Relations with Optical Limit Values

Of the three viewing zones, the intermediate zone measures have more significant relations to the optical limiting values than do measures of the distance and near zones. This is not surprising because two of the three optical limiting values (maximum power rate and minimum zone width) have locations in the intermediate area of the lenses.

There are highly significant positive relations (r values approaching 1.0) between the intermediate zone widths and the minimum zone width on the lenses. The minimum zone width on the lens always occurs in the intermediate zone and therefore will certainly be related to the intermediate zone width; they are essentially the same value. There are also very highly significant positive relations between the intermediate widths and areas and the zone width at the level of maximum power rate change (r values approaching 0.9). Surprisingly, however, the relations between intermediate width and area are not significantly related to zone width at the level of maximum astigmatism (r value range of 0.113–0.308). This is likely a result of the close locations of maximum power rate change and minimum zone width, whereas the location of maximum astigmatism is more variable and is also typically located lower in the lens and closer to the near areas (Table 3).

There is a highly significant negative relation between the maximum power rate and the intermediate widths and areas (r value range from -0.641 to -0.878); that is, lenses with smaller intermediate zones have a higher maximum power rate change. Similar to the situation above with minimum zone width, there is also a strong negative relation between intermediate widths and area and the power rate at the level of the minimum zone width (r value range of -0.585 to -0.822), but not for the power rate at the level of maximum astigmatism (r value range from -0.155 to -0.317).

There is a significant negative relation between maximum astigmatism and the intermediate widths and area (r values range from -0.391 to -0.564). Greater amounts of astigmatism are related to narrower and smaller intermediate zone widths and area. The negative correlations are slightly higher for the amount of astigmatism at the locations of the minimum zone width and maximum power rate (r value range from -0.439 to -0.622), presumably because the maximum amount of astigmatism is located lower on the lens (at the level of the near viewing area), whereas the maximum power rate and minimum zone width are located at the same level as the intermediate viewing zone.

Near Zone Relations with the Optical Limit Values

The near zone measures have numerous significant relations with the maximum power rate and with minimum zone width, but no significant relations with maximum astigmatism. This finding is counter to the suggestion by Simonet et al.⁶ that astigmatism is proportional to the width of the near zone, but their suggestion was based on analysis of only two lenses.

There is a significant positive relation between each near zone

TABLE 2.
Correlation coefficients of optical limit values compared with viewing zones

	Maximum Power Rate					Minimum Zone Width					Maximum Astigmatism				
	Max power rate	Y location	Astig	Zone width	Add pwr	Min Zone Width	Y location	Astig	Add pwr	Pwr rate	Max astig	Y location	Zone width	Add pwr	Pwr rate
Distance															
Width -1															
Pearson r	0.029	0.33	0.502	-0.351	0.13	-0.295	-0.067	0.435	-0.451	0.124	0.463	0.034	0.008	-0.168	-0.015
Significance	0.882	0.086	0.006	0.067	0.508	0.128	0.736	0.021	0.016	0.53	0.013	0.865	0.968	0.392	0.941
Width 0															
Pearson r	0.141	0.373	0.381	-0.327	0.149	-0.25	0.034	0.31	-0.39	0.231	0.317	-0.001	-0.024	-0.227	0.087
Significance	0.473	0.05	0.046	0.09	0.448	0.199	0.864	0.108	0.04	0.238	0.1	0.995	0.905	0.245	0.661
Width 1															
Pearson r	0.149	0.274	0.267	-0.292	0.072	-0.24	0.074	0.216	-0.279	0.2	0.242	0.027	0.069	-0.197	0.052
Significance	0.448	0.158	0.17	0.132	0.717	0.218	0.709	0.269	0.151	0.307	0.215	0.892	0.728	0.316	0.793
Area -1 to +3															
Pearson r	0.111	0.355	0.353	-0.259	0.1	-0.184	0.018	0.285	-0.429	0.177	0.306	0.013	0.032	-0.249	0.069
Significance	0.575	0.064	0.065	0.183	0.611	0.349	0.929	0.141	0.023	0.368	0.114	0.948	0.873	0.201	0.729
Intermediate															
Width 0.75															
Pearson r	-0.647	0.363	-0.501	0.861	0.183	0.867	0.391	-0.551	-0.027	-0.666	-0.549	0.156	0.113	-0.039	-0.155
Significance	<0.001	0.058	0.007	<0.001	0.351	<0.001	0.04	0.002	0.893	<0.001	0.003	0.429	0.566	0.844	0.432
Width 1.00															
Pearson r	-0.671	0.381	-0.519	0.878	0.195	0.902	0.315	-0.566	-0.125	-0.663	-0.544	0.084	0.21	-0.181	-0.152
Significance	<0.001	0.046	0.005	<0.001	0.319	<0.001	0.102	0.002	0.528	<0.001	0.003	0.669	0.284	0.358	0.439
Width 1.25															
Pearson r	-0.641	0.367	-0.547	0.871	0.094	0.927	0.125	-0.622	-0.395	-0.585	-0.564	0.086	0.272	-0.256	-0.165
Significance	<0.001	0.054	0.003	<0.001	0.635	<0.001	0.527	<0.001	0.038	0.001	0.002	0.664	0.162	0.189	0.4
Area 0.75 to 1.50															
Pearson r	-0.878	0.331	-0.439	0.879	-0.021	0.888	0.213	-0.487	-0.396	-0.822	-0.391	0.22	0.308	-0.162	-0.317
Significance	<0.001	0.086	0.019	<0.001	0.916	<0.001	0.277	0.009	0.037	<0.001	0.039	0.26	0.111	0.411	0.1
Near															
First 1.75															
Pearson r	-0.744	0.448	-0.114	0.562	-0.049	0.591	0.264	-0.178	-0.547	-0.648	-0.084	0.228	0.26	-0.257	-0.284
Significance	<0.001	0.017	0.562	0.002	0.804	0.001	0.175	0.366	0.003	<0.001	0.67	0.243	0.181	0.187	0.144
Width 16															
Pearson r	0.615	-0.242	0.215	-0.562	0.131	-0.579	-0.181	0.274	0.365	0.582	0.197	-0.202	-0.247	0.14	0.25
Significance	<0.001	0.215	0.273	0.002	0.508	0.001	0.357	0.158	0.056	0.001	0.316	0.302	0.205	0.477	0.199
Area 16.5															
Pearson r	0.761	-0.443	0.154	-0.563	0.003	-0.564	-0.404	0.19	0.372	0.686	0.13	-0.241	-0.169	0.174	0.277
Significance	<0.001	0.018	0.433	0.002	0.986	0.002	0.033	0.333	0.051	<0.001	0.509	0.216	0.39	0.376	0.153
Width 20															
Pearson r	0.441	-0.105	0.317	-0.472	0.298	-0.487	-0.089	0.344	0.429	0.429	0.34	0	-0.012	0.289	0.038
Significance	0.019	0.593	0.1	0.011	0.124	0.009	0.652	0.073	0.023	0.023	0.077	0.999	0.952	0.136	0.847
Area 20.5															
Pearson r	0.693	-0.324	0.249	-0.586	0.144	-0.594	-0.294	0.29	0.426	0.645	0.23	-0.202	-0.163	0.196	0.235
Significance	<0.001	0.093	0.201	0.001	0.465	0.001	0.128	0.135	0.024	<0.001	0.238	0.302	0.408	0.317	0.228

TABLE 3.

Mean and standard deviations of the optical limit values and locations

	Maximum Power Rate					Minimum Zone Width					Maximum Astigmatism				
	Max power rate	Y location	Astig	Zone width	Add pwr	Min zone Width	Y location	Astig	Add pwr	Pwr rate	Max astig	Y location	Zone width	Add pwr	Pwr rate
Mean	0.15	8.62	1.90	3.90	1.02	3.77	9.20	1.95	1.11	0.14	2.04	12.02	4.82	1.43	0.11
SD	0.02	1.57	0.32	0.59	0.15	0.59	1.00	0.32	0.14	0.02	0.34	3.07	2.23	0.29	0.04

width and area measurement and the maximum power rate (r value range of 0.615–0.761) and also with the maximum power rate at the location of minimum zone width (r value range of 0.429–0.686). Higher power rates are associated with wider and larger near areas.

There are also significant negative relations between near zone widths and areas and minimum zone width (r value range from -0.487 to -0.594) and also with the zone width at the level of maximum power rate change (r value range from -0.472 to -0.586). In other words, lenses with higher power rates and narrower corridors (both of which occur in the intermediate area) have larger near zones.

There are negative relations between near zone areas and the y location of maximum power rate (r value range from -0.443 to -0.324) and the y location of minimum zone width (r value range from -0.404 to -0.294). This means that larger near areas are associated with lenses in which the maximum power rate and the minimum zone width are located higher in the lens. However, these location relations are not nearly as strong as the near zone relations with the magnitudes of power rate and minimum zone width. The near zone relation is based more on the magnitudes of the power rate and the minimum zone width than on their locations.

There is also a positive significant relation between the center power at the location of minimum zone width with near zone widths and areas (r value range from 0.365 to 0.429). A higher power at the minimum zone width location is associated with larger zone widths.

The relations between the three optical limit values and the first occurrence of the +1.75 D add follow exactly those of the near areas and widths, but the relations are opposite in sign because higher first occurrences of +1.75 D add (i.e., a lower y value) are associated with larger near areas.

Interrelations Between the Three Optical Limit Values

The Pearson correlation coefficients and significance levels for these interrelations are shown in Table 4.

There are high correlations of the magnitudes of maximum power rate with minimum zone width ($r = -0.695$, $p < 0.001$) and minimum zone width with maximum astigmatism ($r = -0.616$, $p < 0.001$). A larger minimum zone width is associated with lower magnitude of unwanted astigmatism. A large minimum zone width is also associated with a lower power rate.

However, there is not a significant relation between maximum

power rate and maximum astigmatism ($r = 0.164$, $p = 0.405$), nor is there a significant relation between the maximum power rate with the amount of astigmatism at that location ($r = 0.257$, $p = 0.187$) nor with maximum astigmatism and the value of the power rate at that location ($r = -0.218$, $p = 0.265$).

The locations of maximum power rate and minimum zone width, which are closest to one another on average (Table 3), were significantly related ($r = 0.466$, $p = 0.012$), as were the locations of minimum zone width and maximum astigmatism ($r = 0.405$, $p = 0.032$). The locations of maximum power rate and maximum astigmatism are not significantly related to one another ($r = 0.122$, $p = 0.537$). The maximum power rate change and maximum astigmatism are not significantly related to one another in magnitude or location.

Maximum astigmatism is highly correlated with maximum astigmatism at the locations of maximum power rate ($r = 0.899$, $p < 0.001$) and minimum zone width ($r = 0.919$, $p < 0.001$). Also, maximum astigmatism at the location of maximum power rate is highly correlated with the maximum astigmatism at the location of minimum zone width ($r = 0.971$, $p < 0.001$). The high correlations of unwanted astigmatism at the three locations of the optical limit values are compelling and indicate the magnitude of unwanted astigmatism is fundamental to the design.

The maximum power rate is highly related to the power rate at the location of minimum zone width ($r = 0.908$, $p < 0.001$), but this is expected because those locations are close to one another. The maximum power rate is less strongly related to the power rate at the location of maximum astigmatism ($r = 0.42$, $p = 0.026$), but still significant. The power rates at the locations of maximum astigmatism and minimum zone width are also significantly related ($r = 0.499$, $p = 0.007$). The significant correlations of the power rate magnitude at the three locations indicate that the rate of power change is a basic element of a lens design, but the correlations are not nearly as strong as for unwanted astigmatism. A possible explanation for this is that the power rate can and must change (power rate acceleration) along the corridor, and the manner in which power acceleration changes is an element of particular designs.

The minimum zone width is highly correlated with the zone width at the location of maximum power rate ($r = 0.979$, $p < 0.001$), but this strong relation is almost certainly the result of the proximity of the locations of the minimum zone width and maximum power rate. The minimum zone width is not significantly related to zone width at the location of maximum astigmatism ($r = 0.223$, $p = 0.254$) and the zone widths at the locations of maximum power rate and maximum astigmatism are also not signifi-

TABLE 4.

Correlation coefficients of optical limit values compared with optical limit values

	Maximum Power Rate					Minimum Zone Width					Maximum Astigmatism				
	Max power rate	Y location	Astig	Zone width	Add pwr	Min Zone Width	Y location	Astig	Add pwr	Pwr rate	Max astig	Y location	Zone width	Add pwr	Pwr rate
Maximum Rate															
Max power rate	1	−0.304	0.257	−0.725	−0.009	−0.695	−0.26	0.288	0.304	0.908	0.164	−0.324	−0.341	0.017	0.42
Pearson r		0.116	0.187	<0.001	0.965	<0.001	0.182	0.137	0.115	<0.001	0.405	0.093	0.075	0.931	0.026
Significance															
Y location		1													
Pearson r			0.13	0.268	0.849	0.377	0.466	−0.019	−0.239	−0.138	0.002	0.122	0.171	−0.246	−0.018
Significance			0.509	0.168	<0.001	0.048	0.012	0.924	0.22	0.485	0.993	0.537	0.384	0.206	0.927
Astigmatism			1												
Pearson r				−0.635	0.239	−0.586	−0.192	0.971	−0.135	0.39	0.899	−0.173	−0.221	−0.102	0.145
Significance				<0.001	0.22	0.001	0.329	<0.001	0.494	0.04	<0.001	0.378	0.258	0.606	0.463
Zone width				1											
Pearson r					0.018	0.979	0.313	−0.662	−0.156	−0.746	−0.633	0.112	0.213	−0.167	−0.184
Significance					0.927	<0.001	0.105	<0.001	0.427	<0.001	<0.001	0.571	0.276	0.395	0.348
Add pwr					1										
Pearson r						0.127	0.321	0.105	0.027	0.14	0.07	−0.026	0.019	−0.142	0.132
Significance						0.519	0.095	0.595	0.893	0.476	0.723	0.896	0.924	0.472	0.504
Minimum Zone															
Min Zone Width						1									
Pearson r							0.299	−0.644	−0.221	−0.667	−0.616	0.083	0.223	−0.231	−0.152
Significance							0.122	<0.001	0.258	<0.001	<0.001	0.676	0.254	0.238	0.441
Y location							1								
Pearson r								−0.166	0.566	−0.361	−0.146	0.405	0.124	0.284	−0.259
Significance								0.398	0.002	0.059	0.458	0.032	0.528	0.143	0.183
Astigmatism								1							
Pearson r									−0.02	0.356	0.919	−0.172	−0.269	−0.041	0.128
Significance									0.919	0.063	<0.001	0.382	0.166	0.835	0.515
Add pwr									1						
Pearson r										0.07	−0.069	0.174	−0.118	0.538	−0.104
Significance										0.724	0.729	0.376	0.549	0.003	0.599
Pwr rate										1					
Pearson r											0.233	−0.38	−0.331	−0.126	0.499
Significance											0.234	0.046	0.086	0.525	0.007
Maximum Astigmatism															
Max astigmatism											1				
Pearson r												0.182	0.11	0.214	−0.218
Significance												0.354	0.579	0.273	0.265
Y location												1			
Pearson r													0.841	0.824	−0.878
Significance													<0.001	<0.001	<0.001
Zone width													1		
Pearson r														0.551	−0.875
Significance														0.002	<0.001
Add pwr														1	
Pearson r															−0.709
Significance															<0.001
Pwr rate															1
Pearson r															
Significance															

cantly related ($r = 0.213$, $p = 0.276$). The low correlations of the zone widths with one another at the three locations indicate that the manner in which zone width changes along the corridor is variable across the lens population. In other words, the manner in which zone width changes is an element under the control of the designer.

In general, the relations between the magnitudes of the three optical limit values are much more strongly related to one another than their locations. There appears to be little correlation between the locations and magnitudes of the maximum power rate and the maximum astigmatism.

DISCUSSION

Viewing Zones

Not surprisingly, all of the interzone relations are negative; that is, making one viewing zone wider or larger tends to result in the other two becoming narrower and smaller. By far, the strongest negative relation is between the intermediate and near zones, with r values ranging from approximately -0.42 to -0.80 and being highly statistically significant. The other zone relations are also negative, but the correlation coefficients are not as large (-0.09 to -0.40 for distance and intermediate and -0.25 to -0.43 for distance and near), and statistical significance is not attained for all comparisons. These relations are schematically shown in Fig. 2.

It is noteworthy that each distance width and area measure bears a stronger relation to the near width and area at 20 and 20.5 than to the near width and area at 16 and 16.5. Similarly, the relation to the first occurrence of the $+1.75$ D add is strongest for the distance width at -1 and least for $+1$. These relations show the upper portions of the distance area to be most related to the lower portions of the near zone.

Optical Limit Values

The maximum astigmatism is very highly correlated with the astigmatism at the locations of maximum power rate ($r = 0.899$,

$p < 0.001$) and minimum zone width ($r = 0.919$, $p < 0.001$). Also, the maximum astigmatism at the location of maximum power rate is highly correlated with the maximum astigmatism at the location of minimum zone width ($r = 0.971$, $p < 0.001$). The high correlations of unwanted astigmatism at the three locations of the optical limit values are compelling. The correlations between the locations of the three optical limit values are considerably lower (r values of 0.405, 0.466, and 0.122). The high correlations of the magnitudes of unwanted astigmatism, despite the low correlations of the locations at which it is measured, support the concept that the magnitude of unwanted astigmatism is a fundamental value that co-varies across the locations measured in this study. These findings support use of the maximum amount of astigmatism on the lens as a fundamental measure of the lens design.

Likewise, the maximum power rate is significantly correlated with the power rate at the location of minimum zone width ($r = 0.908$, $p < 0.001$) and at the location of maximum astigmatism ($r = 0.42$, $p = 0.026$). The power rates at the locations of maximum astigmatism and minimum zone width are also significantly related ($r = 0.499$, $p = 0.007$). All the power rate measures are correlated with each other. As with unwanted astigmatism, this indicates that power rate is largely a fundamental value in any particular design.

The minimum width, however, is not nearly as correlated as power rate and astigmatism. The minimum zone width is highly correlated with the zone width at the location of maximum power rate ($r = 0.979$, $p < 0.001$), but not at the location of maximum astigmatism ($r = 0.223$, $p = 0.254$). The zone widths at the locations of maximum power rate and maximum astigmatism are not significantly related ($r = 0.213$, $p = 0.276$). The one significant relation is very strong but is not meaningful by itself because the locations of maximum power rate and minimum zone width are near one another (0.6 mm on average); hence, a correlation of the widths at those two locations would be expected. The width is not well correlated among the locations used in this analysis and acts differently from power rate and unwanted astigmatism. The width does not seem to be a fundamental value in the same way as power rate and unwanted astigmatism.

The relations of the optical limit values to the viewing zones are shown in Fig. 2. Maximum power rate and minimum zone width have strong correlations with the intermediate and near viewing zones but not with the distance viewing zone. Maximum astigmatism has a significant but less strong relation with the intermediate area, none with the near viewing zone, and a significant correlation with the upper width of the distance viewing zone.

Interrelations of Optical Limit Values

Although power rate and unwanted astigmatism each seem a fundamental variable, they are also independent of one another because the correlation between their maximum values is very low ($r = 0.164$, $p = 0.405$), as is the correlation between their locations ($r = 0.122$, $p = 0.537$).

However, both of these fundamental variables have a very significant relation with the minimum zone width. The magnitude of maximum power rate is significantly related to minimum zone width ($r = -0.695$, $p < 0.001$) as are the locations of the two ($r = 0.466$, $p = 0.012$). Likewise, the magnitude of maximum astig-

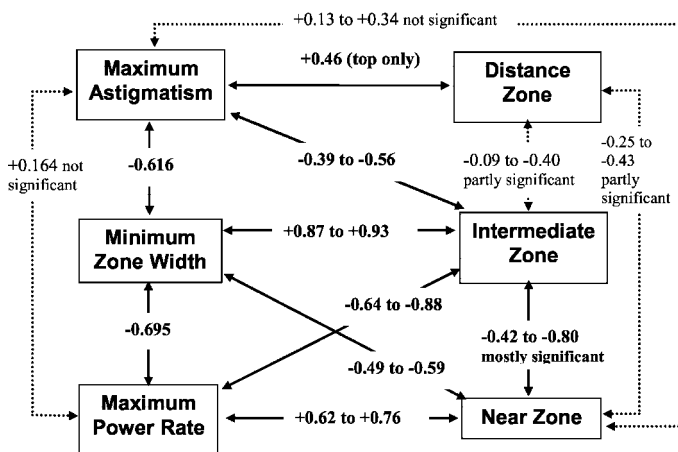


FIGURE 2.

Model of the relations between the optical limit values and viewing zones showing Pearson correlation coefficients. Ranges are indicated because several variables from each viewing zone were tested. Correlations are significant unless indicated otherwise. The most significant relations are in bold.

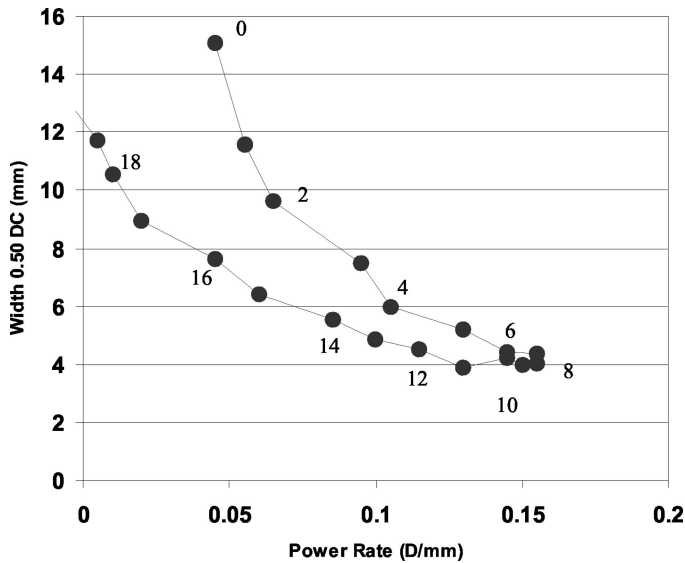


FIGURE 3.

Relation between power rate and zone width (0.50 DC limits) along the corridor from $y = 0$ to $y = 19$ for one selected lens. Alternate y values are indicated next to the data points.

matism is significantly related to minimum zone width ($r = -0.616$, $p < 0.001$) as are their locations ($r = 0.405$, $p = 0.032$). Increases in power rate or astigmatism are each highly correlated with decreased minimum zone width.

The locations and magnitudes of maximum power rate and maximum unwanted astigmatism are independent of one another, whereas the location and magnitude of the minimum zone width is dependent on both.

A model of the relations of the three optical limiting values to one another and to the three viewing zones is shown in Fig. 2.

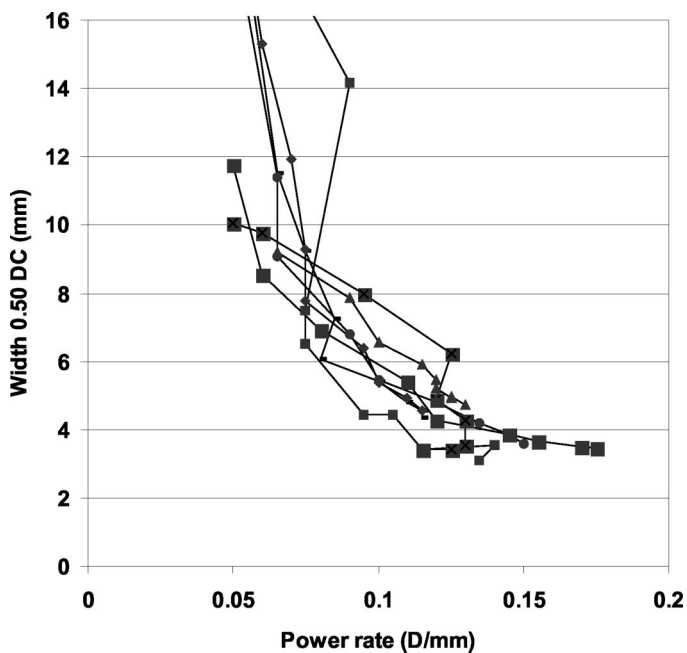


FIGURE 4.

Relation between power rate and zone width (0.50 DC limits) along the corridor from $y = 0$ to $y = 8$. Data are for seven randomly chosen lenses.

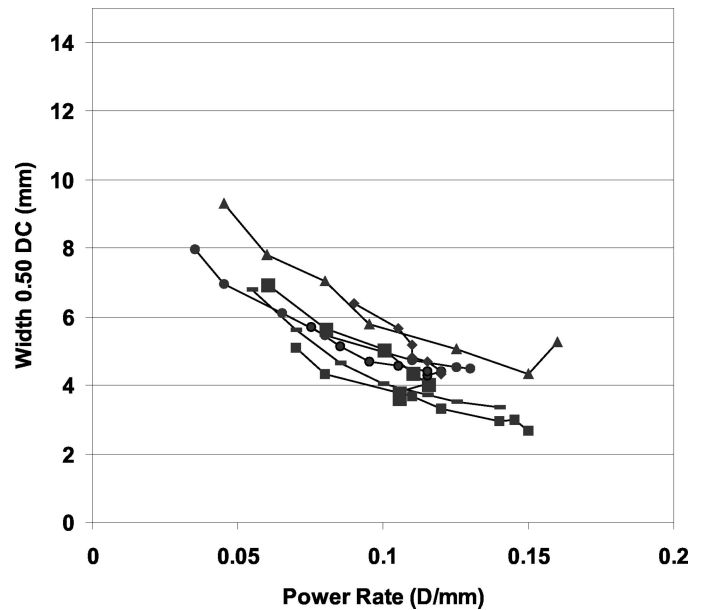


FIGURE 5.

Relation between power rate and zone width (0.50 DC limits) along the corridor from $y = 10$ to $y = 16$. Data are for seven randomly chosen lenses.

Maximum power rate and maximum astigmatism have a significant negative correlation with minimum zone width, but they do not significantly relate to one another. Maximum power rate and minimum zone width have strong correlations with the intermediate and near viewing zones but not with the distance viewing zone. Maximum astigmatism has a significant but less strong relation with the intermediate area, none with the near viewing zone, and a significant correlation with the upper width of the distance viewing zone. The strongest interrelations are among the maximum power rate, minimum zone width, and the intermediate and near viewing zone widths and areas. These two viewing zones are also the two that are most significantly correlated with one another.

The strong relation between maximum power rate and minimum zone width warrants further investigation into the broader relation between power rate and zone width (not just their maximum and minimum values), especially in the intermediate corridor of the lens. The relation between power rate and zone width for a single selected lens is shown in Fig. 3. The points plot in a smooth pattern with contiguous locations in the corridor being contiguous in the plot; this was characteristic of all 28 lenses. The relation between power rate and zone width is different above the location of minimum zone width and maximum power rate than it is below that location, and zone width is greater in relation to the power rate above the minimum and maximum location than below it. This characteristic is also observed with variations in all 28 lenses. The lens selected for Fig. 3 showed the clearest distinction between the relation above and below the minimum and maximum location. The data points for seven randomly selected lenses (every fourth lens was selected from an alphabetical list) for above the minimum and maximum point (chosen as $y = 0$ to $y = 8$) and below the minimum and maximum point (chosen as $y = 10$ to $y = 16$) are shown in Figs. 4 and 5, respectively. The relation between zone width and power rate is quite similar for all lenses.

von Minkwitz⁴ had proposed that unwanted astigmatism at points outside the corridor would grow at twice the rate of increase of power along the corridor. A corollary of his proposal would be that the width of the corridor defined to a given magnitude of astigmatism would be related to the power rate at that location. Clearly, there is a relation between power rate and zone width as shown in Figs. 3 to 5. That the data points for the lenses in Figs. 4 and 5 follow the same general relation indicates a fundamental relation between the power rate magnitude and zone width. However, because the relation is different above and below the minimum and maximum point, it is not perfectly fixed in these lens designs, as von Minkwitz would predict. The relation can be altered, within limits, by individual design.

The significant relation between minimum zone width and maximum astigmatism likewise invites further investigation of the relation between zone width and astigmatism. This relation, using the same lens as in Fig. 3, is shown in Fig. 6. Again, contiguous points in the corridor plot contiguously on the graph and indicate a smooth relation between the two variables along the corridor. Again, the relation above and below the minimum and maximum point is different, with a greater astigmatism-to-width ratio in the lower portion of the lens than in the top. These characteristics are common to all lenses measured in this study as shown by the seven randomly selected lenses graphed in Fig. 7. Within a given lens design, there is a smooth relation between zone width and astigmatism, but the relation is not fixed because it is very different in the upper and lower portions of the corridor for a given lens.

Although there was not a significant correlation between maximum power rate and maximum astigmatism, there is a smooth continuous relation between them for any given lens (Fig. 8). Again, the astigmatism-to-power rate ratio is greater in the lower than the upper portion of the corridor. This relation is not fixed as indicated by the relatively large difference between the upper and lower portions of the corridor and also by the relatively larger difference across lenses (Fig. 9). The relation between power rate

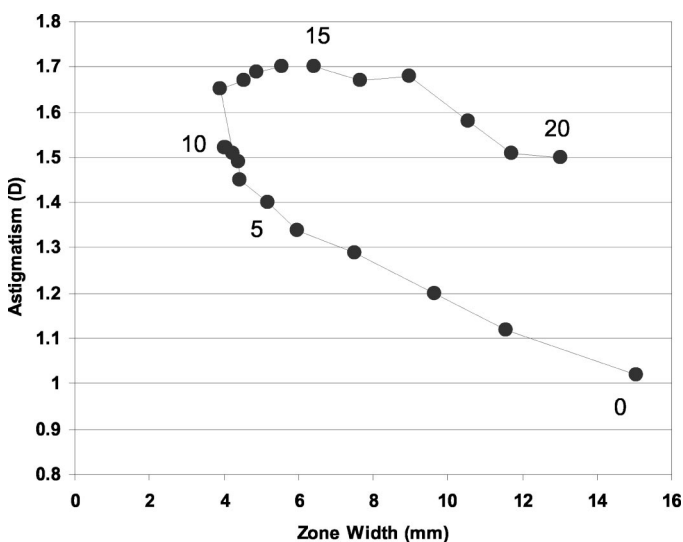


FIGURE 6.

Same lens as in Fig. 3. Relation between zone width (0.50 DC limits) and astigmatism along the corridor from $y = 0$ to $y = 20$. Selected y values are indicated next to the data points.

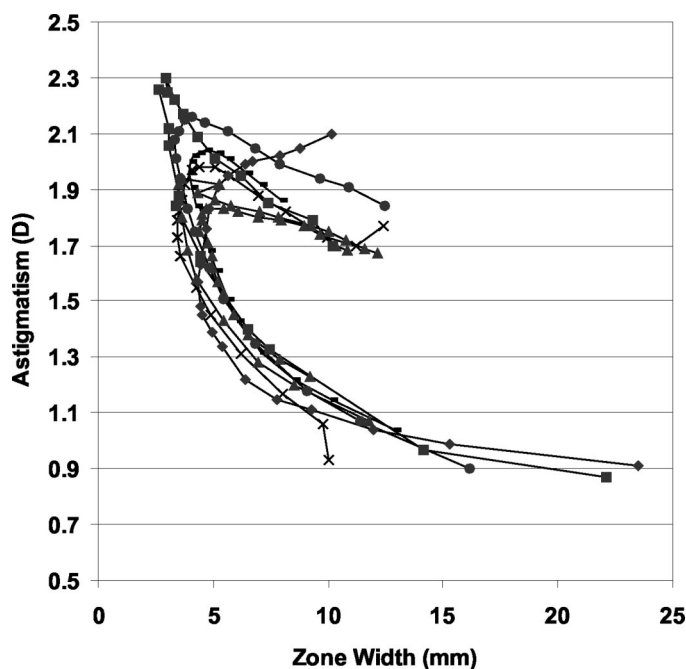


FIGURE 7.

Relation between zone width (0.50 DC limits) and astigmatism along the corridor from $y = 0$ to $y = 20$ for seven randomly chosen lenses.

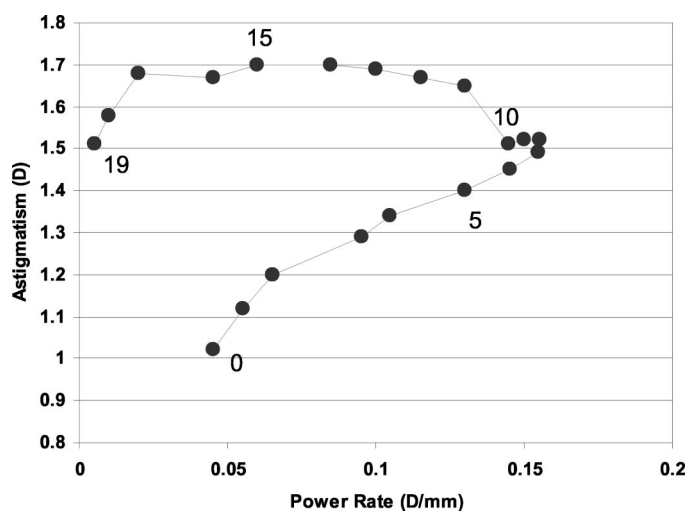


FIGURE 8.

Same lens as in Figs. 3 and 6. Relation between power rate and astigmatism along the corridor from $y = 0$ to $y = 19$. Selected y values are indicated next to the data points.

and astigmatism, as analyzed in this study, is quite different across lenses.

CONCLUSIONS

Power rate and unwanted astigmatism each seem a fundamental value in PAL design. The primary basis for this is the high degree of correlation each variable has with itself at different locations on the lens. Power rate and unwanted astigmatism are also independent of one another, as assessed by the lack of correlation of their maxima with one another across lenses, lack of correlation of the locations

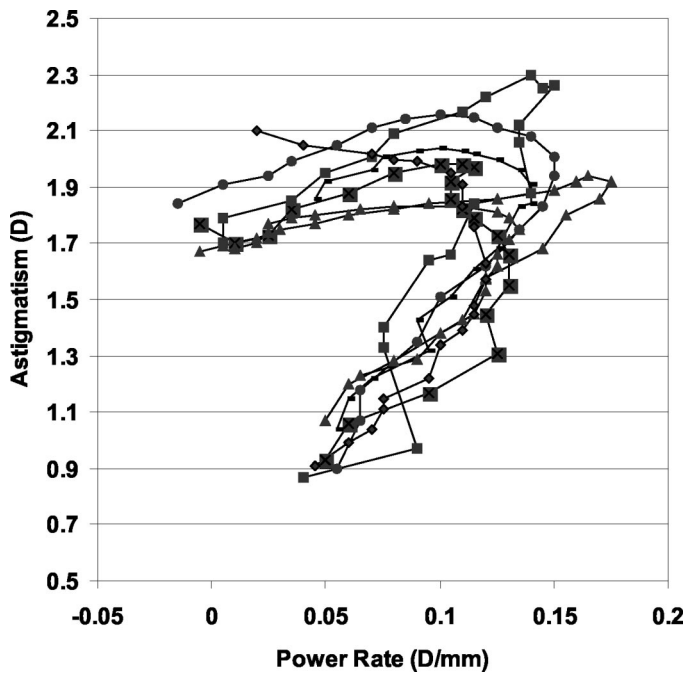


FIGURE 9.

Relation between power rate and astigmatism along the corridor from $y = 0$ to $y = 20$ for seven randomly chosen lenses.

of their maximal values across lenses, and lack of a relation with one another as a function of vertical level across lenses.

Power rate and unwanted astigmatism, however, have a significant relation with zone width. Their maxima are significantly related to minimum zone width across all lenses. The locations of their maxima are both significantly related to the location of minimum zone width, and the relation of each to zone width as a function of vertical level shows significant similarity across lenses.

Zone width differs from power rate and astigmatism in that it does not seem a fundamental value for a given lens; that is, the relations of zone width at different levels is not correlated across lenses. This indicates the designer has discretion to vary zone width at different levels.

The strongest relations are between the power rate, zone width, and the intermediate and near viewing zones. It is likely that the most fundamental optical relation is between power rate and zone width. Astigmatism is related to the intermediate zone and to the upper portion of the distance zone (1 mm above fitting cross), but not related to near viewing zone size.

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